Supplementary Material for "IMU-Based and Auto-Tuned Virtual Decomposition Control for a Hydraulic Rock-Breaker Boom": Proofs of the Stability Results

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Equation, lemma, theorem and section numbers without the prefix S refer to the article.

S-I. Proof of Theorem 1 and Peak Gains

Proof of Theorem 1. (a) By Lemma 1 and the Cauchy–Schwarz inequality, $\frac{d}{dt}\|\tilde{\boldsymbol{x}}\|_{\mathbf{P}} \le -\mu\|\tilde{\boldsymbol{x}}\|_{\mathbf{P}} + \|\boldsymbol{\xi}\|_{\mathbf{P}}$; the comparison lemma [2] yields (10). (b) For a given weight trajectory, (9) is linear; its response is the sum of the response to $\tilde{\boldsymbol{x}}(0)$, bounded by $|\tilde{x}_1| \le \sqrt{2}\|\tilde{\boldsymbol{x}}\|_{\mathbf{P}}$ because $(\mathbf{P}^{-1})_{11} = 2$, and the responses to v, n_ω and η from zero, each bounded by its peak gain, certified below by invariant-ellipsoid inequalities [1]. (c) While $|v - \tilde{x}_1| < \pi$, $\varepsilon = v - \tilde{x}_1$, and (11) gives $|v - \tilde{x}_1| \le \bar{v} + \bar{e} < \pi$, which, by continuity of $\tilde{x}_1$, cannot fail at a first instant. ∎

Peak gains. For an input $d \in \{v, n_\omega, \eta\}$ of (9), with the input vector $\boldsymbol{B}_v(w) = [k_P w, -k_I w]^\top$, $\boldsymbol{B}_n = [1, 0]^\top$ or $\boldsymbol{B}_\eta = [0, -1]^\top$, let $\mathbf{P}_d \succ \mathbf{0}$ and $\alpha_d > 0$ satisfy

$$\mathbf{N}_d := \begin{bmatrix} \mathbf{A}(w)^\top \mathbf{P}_d + \mathbf{P}_d \mathbf{A}(w) + \alpha_d \mathbf{P}_d & \mathbf{P}_d \boldsymbol{B}_d(w) \\ \boldsymbol{B}_d(w)^\top \mathbf{P}_d & -\alpha_d \end{bmatrix} \prec \mathbf{0} \tag{S1}$$

for $w \in \{w_{\min}, 1\}$. Because $\mathbf{N}_d$ is affine in w, (S1) holds on $[w_{\min}, 1]$. Along (9) driven by d alone, $V_d := \tilde{\boldsymbol{x}}^\top \mathbf{P}_d \tilde{\boldsymbol{x}}$ then satisfies $\dot{V}_d \le -\alpha_d(V_d - d^2)$ for every weight trajectory, so that $V_d(t) \le \sup d^2$ from $\tilde{\boldsymbol{x}}(0) = \mathbf{0}$, and $|\tilde{x}_1| \le \gamma_d \sup|d|$ with $\gamma_d := ((\mathbf{P}_d^{-1})_{11})^{1/2}$ [1]. The certificates of Table SI were computed by semidefinite programming, rounded, and verified exactly in rational arithmetic (Sylvester's criterion for $-\mathbf{N}_d$ at both vertices and for $\mathbf{P}_d$); their margins are small, so that the printed digits must be used as given. The response of (9) to the three inputs and to $\tilde{\boldsymbol{x}}(0)$ adds up, which gives (11).

Discretization. For the forward-Euler discretization at $h_s = 1\,\text{ms}$, $\mathbf{F}(w) := (\mathbf{I} + h_s \mathbf{A}(w))^\top \mathbf{P} (\mathbf{I} + h_s \mathbf{A}(w)) - \mathbf{P}$ is a matrix-convex quadratic function of w, because $\mathbf{A}(w)$ is affine in w and $\mathbf{P} \succ \mathbf{0}$; hence $\mathbf{F}(w) \prec \mathbf{0}$ on $[w_{\min}, 1]$ follows from the vertices, where the largest eigenvalues of $\mathbf{F}$ are -3.5×10^{-6} ($w_{\min}$) and -7.9×10^{-4} ($w = 1$).

S-II. Proof of Lemma 2

(a) follows from (12) and (16). (b) For $t \ge h_j$, all commands in (17) follow (12); differentiating (16) and inserting (17) and (a) with $\dot{q}^\circ_{r,j}(t-s) = \dot{q}_{d,j}(t-s) + \lambda_j(t-s)\tilde{e}_j(t-s)$ gives (18).

(c) With constant gains λ_i, as stated in Lemma 2(c), $\dot{q}^\circ_{r,i}$ and $\ddot{q}^\circ_{r,i}$ involve no derivative of λ_i. *Links.* In its body frame at joint i, a planar rigid link obeys Assumption 3(a) with constant $\mathbf{M}_i$ and

$$\mathbf{C}_i(\omega) = m_i^{\mathrm{L}} \omega \begin{bmatrix} 0 & -1 & -c_x \\ 1 & 0 & -c_y \\ c_x & c_y & 0 \end{bmatrix}, \tag{S2}$$

m_i^{L} being the link mass and $[c_x, c_y]$ the center of mass in the body frame; this follows from the Newton–Euler equations in the moving frame and satisfies $\mathbf{C}_i + \mathbf{C}_i^\top = \mathbf{0}$. With $\boldsymbol{F}^{*\circ}_{r,i} := \mathbf{M}_i \dot{\boldsymbol{V}}^\circ_{r,i} + \mathbf{C}_i(\omega_i)\boldsymbol{V}^\circ_{r,i} + \boldsymbol{G}_i(\phi_i) + \mathbf{K}_{V,i}\Delta\boldsymbol{V}_i$, the link dynamics give $\mathbf{M}_i \Delta\dot{\boldsymbol{V}}_i = \boldsymbol{F}^{*\circ}_{r,i} - \boldsymbol{F}^*_i - \boldsymbol{F}_{\mathrm{d},i} - (\mathbf{C}_i + \mathbf{K}_{V,i})\Delta\boldsymbol{V}_i$, so that

$$\dot{\nu}_i = -\Delta\boldsymbol{V}_i^\top \mathbf{K}_{V,i} \Delta\boldsymbol{V}_i + \Delta\boldsymbol{V}_i^\top (\boldsymbol{F}^*_{r,i} - \boldsymbol{F}^*_i) - \Delta\boldsymbol{V}_i^\top (\boldsymbol{\Delta}_i + \boldsymbol{F}_{\mathrm{d},i}) \tag{S3}$$

with $\boldsymbol{\Delta}_i := \boldsymbol{F}^*_{r,i} - \boldsymbol{F}^{*\circ}_{r,i}$, $\boldsymbol{F}^*_{r,i}$ being computed by (13).

Cutting points. The net wrenches are $\boldsymbol{F}^*_i = \boldsymbol{F}_i - \mathbf{U}_{i,i+1}(q_{i+1})\boldsymbol{F}_{i+1}$ and, by (13), $\boldsymbol{F}^*_{r,i} = \boldsymbol{F}_{r,i} - \mathbf{U}_{i,i+1}(\hat{q}_{i+1})\boldsymbol{F}_{r,i+1}$. Since $\boldsymbol{V}^\circ_r$ and $\boldsymbol{V}$ obey the same true kinematics, $\mathbf{U}_{i,i+1}(q_{i+1})^\top \Delta\boldsymbol{V}_i = \Delta\boldsymbol{V}_{i+1} - \boldsymbol{z}\Delta\dot{q}_{i+1}$, $\boldsymbol{z} := [0, 0, 1]^\top$. Hence, for $i = 1, 2$,

$$\begin{aligned} \Delta\boldsymbol{V}_i^\top (\boldsymbol{F}^*_{r,i} - \boldsymbol{F}^*_i) &= p_i - p_{i+1} + \Delta\dot{q}_{i+1}(\mathcal{T}_{r,i+1} - \mathcal{T}_{i+1}) \\ &\quad - \Delta\boldsymbol{V}_i^\top \big[\mathbf{U}_{i,i+1}(\hat{q}_{i+1}) - \mathbf{U}_{i,i+1}(q_{i+1})\big] \boldsymbol{F}_{r,i+1}, \end{aligned} \tag{S4}$$

with the joint torques $\mathcal{T}_{r,i} := \boldsymbol{z}^\top \boldsymbol{F}_{r,i} = m_i(\hat{q}_i) f_{r,i}$ and $\mathcal{T}_i := \boldsymbol{z}^\top \boldsymbol{F}_i = m_i(q_i) f_i$, and $p_3 - p_4$ for $i = 3$, the tool frame being fixed on link 3. With the turret fixed in the boom plane, $\Delta\boldsymbol{V}_1 = \boldsymbol{z}\Delta\dot{q}_1$ and $p_1 = \Delta\dot{q}_1(\mathcal{T}_{r,1} - \mathcal{T}_1)$. Summing (S3) over i telescopes the VPFs and leaves $\sum_i \Delta\dot{q}_i(\mathcal{T}_{r,i} - \mathcal{T}_i) - p_4$, where

$$\Delta\dot{q}_i(\mathcal{T}_{r,i} - \mathcal{T}_i) = (\dot{x}^\circ_{r,i} - \dot{x}_i)\Delta f_i + \Delta\dot{q}_i\big(m_i(\hat{q}_i) - m_i(q_i)\big) f_{r,i}. \tag{S5}$$

Actuators. By Assumption 3(b), $\Delta\dot{f}_i = -\mathcal{B}k_{f,i}\Delta f_i - \mathcal{B}k_{x,i}(\dot{x}_{r,i} - \hat{\dot{x}}_i) - \mathcal{B}[c_{p,i}(\hat{x}_i)\hat{\dot{x}}_i - c_{p,i}(x_i)\dot{x}_i]$, so that

$$\dot{\nu}_{\mathrm{a},i} = -\tfrac{k_{f,i}}{k_{x,i}}\Delta f_i^2 - (\dot{x}^\circ_{r,i} - \dot{x}_i)\Delta f_i + \Delta f_i \Delta_{\mathrm{a},i} \tag{S6}$$

with $\Delta_{\mathrm{a},i} := \dot{x}^\circ_{r,i} - \dot{x}_{r,i} + \hat{\dot{x}}_i - \dot{x}_i - [c_{p,i}(\hat{x}_i)\hat{\dot{x}}_i - c_{p,i}(x_i)\dot{x}_i]/k_{x,i}$. Adding the sums cancels the actuator VPFs $(\dot{x}^\circ_{r,i} - \dot{x}_i)\Delta f_i$ and gives (19) and (20).

Bound. On $\mathcal{S}$, $\dot{q}$, $\dot{q}^\circ_r$, $\ddot{q}^\circ_r$, $f_{r,i}$ and $\boldsymbol{F}_{r,i}$ are bounded, and m_i, $c_{p,i}$, $L_{c,i}$, $\mathbf{U}_{i,i+1}$ and the VDC recursions are smooth. Now $\dot{x}_{r,i} - \dot{x}^\circ_{r,i} = (m_i(\hat{q}_i) - m_i(q_i))\dot{q}^\circ_{r,i} - m_i(\hat{q}_i)\lambda_i e_{q,i}$, $\hat{\dot{x}}_i - \dot{x}_i = m_i(\hat{q}_i)e_{\omega,i} + (m_i(\hat{q}_i) - m_i(q_i))\dot{q}_i$, and

TABLE SI
CERTIFICATES (S1) OF THE PEAK GAINS FOR $k_P = 0.8\,\mathrm{s}^{-1}$, $k_I = 0.01\,\mathrm{s}^{-2}$, $w_{\min} = 0.02$

d	$10^3\alpha_d$	p_{11}	p_{12}	p_{22}	$\gamma_d \le$	frozen
v	7.9433	3.349208×10^{-2}	1.3527258	115.20053	7.54	1.6/1.03
n_ω	5.0119	5.0882495×10^{-5}	1.5385545×10^{-3}	0.15599101	168	82/2.4
η	6.3096	3.9063174×10^{-8}	1.540013×10^{-6}	1.2031021×10^{-4}	7189	6311/100

$\mathbf{P}_d = [p_{ij}]$ in units for which $\tilde{x}_1$ is in rad, $\tilde{x}_2$ in rad/s and d in its own unit; α_d in s^{-1}; γ_n in s, γ_η in s^2. Last column: exact peak gains for the frozen weights $w_{\min}$ and 1, which are lower bounds of γ_d.

$\|\mathbf{U}_{i,i+1}(\hat{q}) - \mathbf{U}_{i,i+1}(q)\| \le \|\mathbf{T}_i\||\hat{q} - q|$, because $\mathbf{U}_{i,i+1}(q) = \mathbf{T}_i\,\mathrm{diag}(\mathbf{R}(q), 1)$ with a constant translation $\mathbf{T}_i$ and $\|\mathbf{R}(\hat{q}) - \mathbf{R}(q)\| = 2|\sin\frac{\hat{q}-q}{2}|$. The computed and the true required wrench of link i are the same smooth function of $(\hat{q}, \hat{\dot{q}}, \dot{q}_r, \ddot{q}_r)$ and $(q, \dot{q}, \dot{q}_r^\circ, \ddot{q}_r^\circ)$, whose arguments differ by $\boldsymbol{e}_q$, $\boldsymbol{e}_\omega$, $-\lambda\boldsymbol{e}_q$ and $-\lambda\boldsymbol{e}_\omega$ (Assumption 3(c)); by the mean-value theorem, $\|\boldsymbol{\Delta}_i\|$ and $|\Delta_{\mathrm{a},i}|$ are bounded by multiples of $\|\boldsymbol{e}_q\| + \|\boldsymbol{e}_\omega\|$. Finally, $|\Delta\dot{q}_i| \le (1 + \|\mathbf{U}_{i-1,i}\|)\|\boldsymbol{\chi}\|$, which gives the bound on ϱ and $\|\Delta\dot{\boldsymbol{q}}\| \le c_J\|\boldsymbol{\chi}\|$ for some $c_J > 0$. ■

S-III. DELAY LOOP: PROOF OF LEMMA 3 AND KERNELS WITH TAILS

Proof of Lemma 3. For the given a and $\mathcal{K}_t$, (21) is linear; write $y = y_h + y_u$, where y_u vanishes on $[t_0 - 2h, t_0 - h]$ and solves (21), and y_h solves (21) with $u = 0$ for $t \ge t_0 - h$. Substituting $y(t-s) = y(t) - \int_{t-s}^{t}\dot{y}(r)\,dr$, valid for y_h at $t \ge t_0$ and, since y_u is absolutely continuous on $[t_0 - 2h, \infty)$, for y_u at $t \ge t_0 - h$,

$$\dot{y}(t) = -\int a(t,s)\,\mathcal{K}_t(ds)\,y(t) + \int a(t,s)\int_{t-s}^{t}\dot{y}(r)\,dr\,\mathcal{K}_t(ds) + u(t). \quad \text{(S7)}$$

Homogeneous part. Let $0 < \mu' < \mu_{\mathrm{d}}$ and $E := e^{2\mu' h}$, so that $\vartheta E < 1$ and $\mu' < a_{\mathrm{lo}}(1 - \vartheta E)$. Let $z(t) := e^{\mu'(t-t_0)}|y_h(t)|$ and $Y_0 := \|y_h\|_{[t_0-2h,t_0]}$; then $z \le Y_0$ on $[t_0 - 2h, t_0]$. Suppose that $z(t) > Y_0 + \epsilon$ for some t and $\epsilon > 0$, and let $t_1 := \inf\{t \ge t_0 : z(t) > Y_0 + \epsilon\}$, so that $z(t_1) = Y_0 + \epsilon$ and $D^+z(t_1) \ge 0$. Then $z(r) \le z(t_1)$ for $r \le t_1$, i.e., $|y_h(r)| \le E|y_h(t_1)|$ on $[t_1 - 2h, t_1]$, so that $|\dot{y}_h(r)| \le a_{\mathrm{hi}}E|y_h(t_1)|$ on $[t_1 - h, t_1]$ by (21). In (S7), evaluated with right limits at t_1, this gives

$$D^+|y_h(t_1)| \le -\int a(t_1,s)(1 - a_{\mathrm{hi}}Es)\,\mathcal{K}_{t_1}(ds)\,|y_h(t_1)| \le -a_{\mathrm{lo}}(1 - \vartheta E)|y_h(t_1)|,$$

since $1 - a_{\mathrm{hi}}Es \ge 1 - \vartheta E > 0$ for $s \le h$. Hence $D^+z(t_1) \le (\mu' - a_{\mathrm{lo}}(1 - \vartheta E))z(t_1) < 0$, which contradicts the definition of t_1. Thus $|y_h(t)| \le e^{-\mu'(t-t_0)}Y_0$ for every $\mu' < \mu_{\mathrm{d}}$, and hence for μ_{d}.

Forced part. Let $U := \|u\|_{[t_0-h,t]}$, suppose that $|y_u|$ exceeds $\Gamma U + \epsilon$ somewhere on $[t_0 - h, t]$, and let t_1 be the infimum of such instants, so that $|y_u(t_1)| = \Gamma U + \epsilon$ and $D^+|y_u(t_1)| \ge 0$. Then $|y_u(r)| \le |y_u(t_1)|$ for $r \le t_1$, $|\dot{y}_u(r)| \le a_{\mathrm{hi}}|y_u(t_1)| + U$ on $[t_1 - h, t_1]$, and (S7) gives $D^+|y_u(t_1)| \le -a_{\mathrm{lo}}(1 - \vartheta)|y_u(t_1)| + (1 + \vartheta)U = -a_{\mathrm{lo}}(1 - \vartheta)\epsilon < 0$, a contradiction. Thus $|y_u| \le \Gamma U$.

Conclusion. $\|y_h\|_{[t_0-2h,t_0]} \le \|y\|_{[t_0-2h,t_0]} + \Gamma\|u\|_{[t_0-h,t_0]}$ gives (22). Restarting at $t' \ge t_0$ gives $|y(t)| \le e^{-\mu_{\mathrm{d}}(t-t')}(\|y\|_{[t'-2h,t']} + \Gamma U') + \Gamma U'$, $U' := \sup_{s \ge t'-h}|u(s)|$, and letting $t \to \infty$, then $t' \to \infty$, gives the asymptotic gain. ■

Lemma S1 (Kernels with tails)**.** *Let $\mathcal{K}_t$ be probability measures on $[0, \infty)$ with $\mathcal{K}_t((h, \infty)) \le \bar{\epsilon}$ and $\sup_t \mathcal{K}_t((s, \infty)) \to 0$ as $s \to \infty$, let a and u be as in Lemma 3, and let y be bounded and measurable on $(-\infty, t_0]$, absolutely continuous on $[t_0, \infty)$, and satisfy (21), with the integral taken over $[0, \infty)$, for $t \ge t_0$. If $\vartheta = a_{\mathrm{hi}}h < 1$ and $2\Gamma' a_{\mathrm{hi}}\bar{\epsilon} < 1$, $\Gamma' := (1 + \vartheta)/(a_{\mathrm{lo}}(1 - \bar{\epsilon})(1 - \vartheta))$, then*

$$\sup_{t \ge t_0}|y(t)| \le \frac{\mathrm{e}\|y\|_{(-\infty,t_0]} + (\mathrm{e}h + 2\Gamma')\|u\|_{[t_0,\infty)}}{1 - 2\Gamma' a_{\mathrm{hi}}\bar{\epsilon}} \quad \text{(S8)}$$

and $\limsup|y| \le \Gamma' \limsup|u|/(1 - \Gamma' a_{\mathrm{hi}}\bar{\epsilon})$.

Proof. Let $Y(t) := \sup_{r \le t}|y(r)|$. On $[t_0, t_0 + h]$, $|\dot{y}| \le a_{\mathrm{hi}}Y + |u|$, and Gronwall's inequality gives $Y(t_0 + h) \le e^{\vartheta}(Y(t_0) + h\|u\|) \le \mathrm{e}(Y(t_0) + h\|u\|)$. For $t \ge t_0$, (21) is exactly

$$\dot{y}(t) = -\int_{[0,h]} \bar{a}(t,s)\,y(t-s)\,\bar{\mathcal{K}}_t(ds) + \bar{u}(t),$$

with the probability measure $\bar{\mathcal{K}}_t := \mathcal{K}_t|_{[0,h]}/\mathcal{K}_t([0,h])$, $\bar{a} := a\,\mathcal{K}_t([0,h]) \in [a_{\mathrm{lo}}(1 - \bar{\epsilon}), a_{\mathrm{hi}}]$ and $\bar{u} := u - \int_{(h,\infty)} a(t,s)y(t-s)\mathcal{K}_t(ds)$, $|\bar{u}(t)| \le |u(t)| + a_{\mathrm{hi}}\bar{\epsilon}Y(t)$. Lemma 3, applied from $t_0 + h$ with the constant Γ', gives $|y(t)| \le Y(t_0 + h) + 2\Gamma'(\|u\| + a_{\mathrm{hi}}\bar{\epsilon}Y(t))$ for $t \ge t_0 + h$; hence $Y(t)(1 - 2\Gamma' a_{\mathrm{hi}}\bar{\epsilon}) \le Y(t_0 + h) + 2\Gamma'\|u\|$, which is (S8). The asymptotic gain of Lemma 3 gives $\limsup|y| \le \Gamma' \limsup|\bar{u}|$, and $\limsup|\bar{u}| \le \limsup|u| + a_{\mathrm{hi}}\bar{\epsilon}\limsup|y|$, because the mass of $\mathcal{K}_t$ on (s, ∞) vanishes uniformly as $s \to \infty$ and y is bounded. ■

Delay followed by a lag. For a delay τ followed by a first-order lag T, the kernel has the density $T^{-1}e^{-(s-\tau)/T}$ on $[\tau, \infty)$, so that $\bar{\epsilon} = e^{-(h-\tau)/T}$ for a truncation point $h \ge \tau$. With $\tau + T$ equal to the identified delay τ_j, $a_{\mathrm{hi}} = \zeta_{\mathrm{hi}}\bar{\lambda}_j$ and $a_{\mathrm{lo}} = \zeta_{\mathrm{lo}}\lambda_{\mathrm{lo}}$ (or $\zeta_{\mathrm{lo}}\bar{\lambda}_j$ for a constant λ_j), the largest lag share T/τ_j for which some $h \in [\tau, 1/a_{\mathrm{hi}})$ satisfies $2\Gamma' a_{\mathrm{hi}}\bar{\epsilon} < 1$ was found by bisection over the share and a search over h. For the delays of Table I, it is 0.45–0.49 for the boom joints and 0.66–0.67 for the slew with $\zeta_j \equiv 1$, 0.73 with a constant λ_j, and 0.29–0.32 and 0.39–0.40 with $\zeta_j \in [0.8, 1.2]$ (0.43 with a constant λ_j). Commands issued before the loop is closed, replaced by the feedback law extended to $t < 0$, enter u as a bounded input that vanishes, because $\mathcal{K}_t((t, \infty)) \to 0$.

S-IV. Proofs of Theorems 2 and 3 and of Remark 1

Theorem 2. (i) is Theorem 1(b), (c) under Assumption 1; (9) contains neither $\tilde{e}_j$ nor any actuator state. (ii) By Lemma 2(b), $y=\tilde{e}_j$ satisfies (21) for $t\ge h_j$ with $a=a_j\in[\zeta_{\rm lo}\lambda_{\rm lo},\zeta_{\rm hi}\bar{\lambda}_j]$, $\mathcal{K}_t=\mathcal{K}_{j,t}$, $h=h_j$ and $u=\int a_j e_{q,j}(t-s)\mathcal{K}_{j,t}(ds)+\varpi_j$, so that $|u(t)|\le\zeta_{\rm hi}\bar{\lambda}_j\|e_{q,j}\|_{[t-h_j,t]}+|\varpi_j(t)|$; (23) is $a_{\rm hi}h<1$, and Lemma 3 with $t_0=2h_j$ gives (24) and the asymptotic gain. (iii) For $t\ge 8h_j$, apply (24) on $[t/2,t]$ and on $[2h_j,t/2]$, and bound $\|e_{q,j}\|_{[t/2-2h_j,t]}\le\sqrt{2}e^{-\mu(t/2-2h_j)}\|\tilde{\boldsymbol{x}}_j(0)\|_{\mathbf{P}}+\bar{e}_{\infty,j}$ and $\|e_{q,j}\|_{[0,t/2]}\le\sqrt{2}\|\tilde{\boldsymbol{x}}_j(0)\|_{\mathbf{P}}+\bar{e}_{\infty,j}$ by (11), $\bar{e}_{\infty,j}:=\gamma_v\bar{v}_j+\gamma_n\bar{n}_j+\gamma_\eta\bar{\eta}_j$; this gives (25), in which the initial errors decay with $\bar{\mu}_j$, and for $2h_j\le t<8h_j$, (24) suffices. With ideal sensors and $\varpi_j\equiv 0$, (25) and (10) give exponential convergence. (iv) $e_{q,0}$ integrates the slew-rate error, also while the angle is held, and (ii) holds for $j=0$ with this input. A reset of the reference at a segment start t_s changes $\tilde{e}_j$ at $t_s+\tau_j$ by at most the error at the end of the previous segment; (ii) then holds again from $t_s+\tau_j+h_j$, because Lemma 3 needs only a bounded history. ∎

Theorem 3. Tracking errors. Since $\dot{q}=\dot{q}_r^\circ-\Delta\dot{q}$, $\dot{\tilde{\boldsymbol{e}}}=-\boldsymbol{\Lambda}\tilde{\boldsymbol{e}}+\Delta\dot{\boldsymbol{q}}+\boldsymbol{d}_q$, $\boldsymbol{d}_q:=\dot{\boldsymbol{q}}_d(t-\boldsymbol{\tau})-\dot{\boldsymbol{q}}_d(t)$, so that $\nu_0:=\frac{1}{2}\|\tilde{\boldsymbol{e}}\|^2$ satisfies $\dot{\nu}_0\le-\lambda_{\rm m}\|\tilde{\boldsymbol{e}}\|^2+\|\tilde{\boldsymbol{e}}\|(c_J\|\boldsymbol{\chi}\|+\|\boldsymbol{d}_q\|)$, $\lambda_{\rm m}:=\min_i\lambda_i$, with c_J from Section S-II.

Links and actuators. Since the low-pass is linear, $\boldsymbol{e}_\omega=\boldsymbol{e}_f+\boldsymbol{d}_f$. By Lemma 2(c), with $a_\chi:=\min_i\{\lambda_{\min}(\mathbf{K}_{V,i}),k_{f,i}/k_{x,i}\}$ and $|p_4|\le c_4\|\boldsymbol{\chi}\|\|\boldsymbol{F}_{\rm tip}\|$,

$$\textstyle\sum_i(\dot{\nu}_i+\dot{\nu}_{{\rm a},i})\le-a_\chi\|\boldsymbol{\chi}\|^2+b_\varrho\|\boldsymbol{\chi}\|(\|\boldsymbol{e}_q\|+\|\boldsymbol{e}_f\|+\|\boldsymbol{d}_f\|)+\|\boldsymbol{\chi}\|(\|\boldsymbol{F}_{\rm d}\|+c_4\|\boldsymbol{F}_{\rm tip}\|).\quad\text{(S9)}$$

Rate-filter lag. On $\mathcal{S}$, the joint-space form of the closed loop, $\mathbf{M}(q)\ddot{\boldsymbol{q}}=m(q)\circ(f_r-\Delta f)-\mathbf{C}\dot{\boldsymbol{q}}-\boldsymbol{g}(q)+\boldsymbol{\tau}_{\rm d}$ with the required torques $\mathcal{T}_{r,i}$ of the VDC recursion, gives $\|\ddot{\boldsymbol{q}}\|\le\|\ddot{\boldsymbol{q}}_d\|+L(\|\tilde{\boldsymbol{e}}\|+\|\boldsymbol{\chi}\|+\|\boldsymbol{e}_q\|+\|\boldsymbol{e}_f\|+\|\boldsymbol{d}_f\|+\|\boldsymbol{d}_{\rm c}\|)$ for some $L>0$. Hence $\frac{d}{dt}\frac{1}{2}\|\boldsymbol{d}_f\|^2=-\|\boldsymbol{d}_f\|^2/T_v-\boldsymbol{d}_f^\top\ddot{\boldsymbol{q}}$ is at most $-(1/T_v-L)\|\boldsymbol{d}_f\|^2+\|\boldsymbol{d}_f\|\big(L(\|\tilde{\boldsymbol{e}}\|+\|\boldsymbol{\chi}\|+\|\boldsymbol{e}_q\|+\|\boldsymbol{e}_f\|+\|\boldsymbol{d}_{\rm c}\|)+\|\ddot{\boldsymbol{q}}_d\|\big)$.

Cascade estimate. With $k_e:=a_\chi\lambda_{\rm m}/(2c_J^2)$, Young's inequality gives $k_ec_J\|\tilde{\boldsymbol{e}}\|\|\boldsymbol{\chi}\|\le\frac{k_e\lambda_{\rm m}}{2}\|\tilde{\boldsymbol{e}}\|^2+\frac{a_\chi}{4}\|\boldsymbol{\chi}\|^2$, so that $\boldsymbol{v}_0=(\tilde{\boldsymbol{e}},\boldsymbol{\chi})$ is dissipated at the rate $a_v:=\min\{k_e\lambda_{\rm m}/2,3a_\chi/4\}$. The couplings with $\boldsymbol{d}_f$ satisfy $b_\varrho\|\boldsymbol{\chi}\|\|\boldsymbol{d}_f\|+L\|\boldsymbol{d}_f\|(\|\tilde{\boldsymbol{e}}\|+\|\boldsymbol{\chi}\|)\le\frac{a_v}{4}\|\boldsymbol{v}_0\|^2+\frac{2b_\varrho^2+4L^2}{a_v}\|\boldsymbol{d}_f\|^2$, and the dissipation $-(1/T_v-L)\|\boldsymbol{d}_f\|^2$ dominates the last term with the margin $\|\boldsymbol{d}_f\|^2/(2T_v)$ if $T_v\le T_v^*:=\big(2L+2(2b_\varrho^2+4L^2)/a_v\big)^{-1}$. Young's inequality applied to the remaining terms gives

$$\dot{W}_{\rm c}\le-\tfrac{a_v}{2}\|\boldsymbol{v}_0\|^2-\tfrac{1}{4T_v}\|\boldsymbol{d}_f\|^2+\gamma'_W(\|\boldsymbol{e}_q\|^2+\|\boldsymbol{e}_f\|^2)+\gamma'\|\boldsymbol{d}_{\rm c}\|^2.$$

As $W_{\rm c}$ is bounded by a quadratic form in $(\tilde{\boldsymbol{e}},\boldsymbol{\chi},\boldsymbol{d}_f)$, (27) follows, and the comparison lemma [2] gives

$$W_{\rm c}(t)\le e^{-\alpha_W t}W_{\rm c}(0)+\gamma_W\int_0^t e^{-\alpha_W(t-s)}\big(\|\boldsymbol{e}_q\|^2+\|\boldsymbol{e}_f\|^2+\|\boldsymbol{d}_{\rm c}\|^2\big)(s)\,ds\quad\text{(S10)}$$

as long as the trajectory stays in $\mathcal{S}$. The observer errors obey Theorem 1, which does not involve $(\tilde{\boldsymbol{e}},\boldsymbol{\chi})$; $|\tilde{x}_{2,i}|\le 2(k_I/k_P)\|\tilde{\boldsymbol{x}}_i\|_{\mathbf{P}}$ because $(\mathbf{P}^{-1})_{22}=4k_I^2/k_P^2$, and $e_{f,i}$ is a low-pass of $-\tilde{x}_{2,i}+n_{\omega,i}$ with unit gain, so that $|e_{f,i}|\le\sup(2\frac{k_I}{k_P}\|\tilde{\boldsymbol{x}}_i\|_{\mathbf{P}}+|n_{\omega,i}|)$ up to a decaying term. With ideal sensors and $\boldsymbol{d}_{\rm c}\equiv\mathbf{0}$, $\|\boldsymbol{e}_q\|^2+\|\boldsymbol{e}_f\|^2$ decays exponentially, and so does $W_{\rm c}$; then $\Delta\boldsymbol{V}_i,\Delta f_i\in\mathcal{L}_2\cap\mathcal{L}_\infty$, as in the VDC theorem [3], and the convergence is exponential. ∎

Remark 1. Let $V_{\rm o}:=\sum_i(\tilde{\boldsymbol{x}}_i^\top\mathbf{P}\tilde{\boldsymbol{x}}_i+\frac{\lambda_QT_v}{4}e_{f,i}^2)$ with $\lambda_Q:=\lambda_{\min}(\mathbf{Q}(w_{\min}))$. By Lemma 1, $\frac{d}{dt}\sum_i\tilde{\boldsymbol{x}}_i^\top\mathbf{P}\tilde{\boldsymbol{x}}_i\le-\lambda_Q\|\tilde{\boldsymbol{x}}\|^2+2\|\mathbf{P}\|\|\tilde{\boldsymbol{x}}\|\|\boldsymbol{\xi}\|$, and $T_v\dot{e}_{f,i}=-\tilde{x}_{2,i}+n_{\omega,i}-e_{f,i}$ gives $\frac{d}{dt}\frac{\lambda_QT_v}{4}e_{f,i}^2\le-\frac{\lambda_Q}{4}e_{f,i}^2+\frac{\lambda_Q}{2}(\tilde{x}_{2,i}^2+n_{\omega,i}^2)$. Hence $\dot{V}_{\rm o}\le-\frac{\lambda_Q}{4}\|\tilde{\boldsymbol{x}}_{\rm e}\|^2+2\|\mathbf{P}\|\|\tilde{\boldsymbol{x}}\|\|\boldsymbol{\xi}\|+\frac{\lambda_Q}{2}\|\boldsymbol{n}_\omega\|^2$, which contains no control error, i.e., $\alpha_2=\lambda_Q/4$ and $\gamma_{\rm o}=0$. Before Young's inequality is applied to the cross terms with $\boldsymbol{e}_q$ and $\boldsymbol{e}_f$, the proof of Theorem 3 gives $\dot{W}_{\rm c}\le-\alpha_1\|\boldsymbol{v}\|^2+\gamma_{\rm c}\|\boldsymbol{v}\|\|\tilde{\boldsymbol{x}}_{\rm e}\|$ up to disturbance terms, with $\alpha_1=\min\{a_v/2,1/(4T_v)\}$ and $\gamma_{\rm c}$ collecting the coefficients of these cross terms ($\|\boldsymbol{e}_q\|\le\|\tilde{\boldsymbol{x}}\|$). For $\dot{V}_{\rm c}\le-a\|z\|^2+b\|z\|\|e\|$ and $\dot{V}_{\rm o}\le-g\|e\|^2+r\|e\|\|z\|$, the composite function $V_{\rm c}+cV_{\rm o}$ decreases if $4acg>(b+cr)^2$; maximizing $4acg-(b+cr)^2$ over $c>0$ shows that a suitable c exists if and only if $ag>br$, which holds for every b if $r=0$. ∎

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